Chapter 2.4 -- Computers: The grid's distributed brain

An octopus has roughly five hundred million neurons. That alone makes it a respectable thinker, somewhere between a domestic cat and a grey parrot. But what's even more remarkable is where this brain lives. Two-thirds of an octopus's brain is not inside the octopus's head. It's distributed along its eight arms -- about three hundred and thirty million little processors in suckers and skin and connective tissue. When an octopus opens a jar -- and they do open jars, in laboratory tanks all over the world -- the central brain decides it would like the snack inside. It does not micromanage the rest. Each arm reaches independently. It's gripping, tasting, twisting, and autonomously problem-solving. With an octopus this surprises us because it's distributed the actual brain cells that we associate with intelligence, but your own intelligence is also less concentrated in your brain than you might think.

Your intestines take care of the highly complex food processing with no instructions from your brain whatsoever, and your brain wouldn't be able to replace them, not even with a PhD in food chemistry. Your heart has an autonomous build in pacemaker. Your spinal chord does not just transmit signals but also helps you to keep your balance. The cells in your body are highly autonomous advanced micro-factories that do their thing without coordination from your brain and with only minimal communication with the outside world. You have a microbiome consisting of trillions of guests with their own opinions. These bacteria influence your immune reactions, hunger, anxiety and even your risk taking behaviour. So your perceive yourself as one integrated personality, but you are a federation of countless semi-autonomous parts.

I say all this, because the grid will also get such a distributed brain. Just like you or the octopus. So far we've been wrongly trying to pretend the grid is a monolith that should be guided from the top down, but that is entirely the wrong way to go about it. Instead every appliance will use a little computer to become a highly competent energy planner, and together they will make the grid amazingly resilient and cost-effective at the same time.

Moore's law

I hope you are not yet sick and tired of learning curves dear reader, because it's impossible to talk about computers without discovering their learning curve is the craziest of all. I'm betting it's even the first learning curve you heard about: Moore's law.

It was named after Gordon Moore, the Intel co-founder who in 1965 made a casual observation in a trade magazine and ended up describing the central economic fact of the late twentieth century. Moore noticed that the number of transistors you could fit onto a chip was doubling roughly every two years. He projected the trend forward for ten years. If we translate is more broadly in computing per dollar, it has held, along many announcements of its impending death, for sixty years. It even turns out this predictable increase in compute per dollar has been going on for 200 years.

As the transistor count per chip increased, the price per computation continued to fall. For sixty years, computing power has become thirty to forty percent cheaper every year. Every year, decade after decade. The cost of a single arithmetic operation has fallen by something like twelve orders of magnitude. That's a *trillion* times more computing per dollar, with GPU's going faster still. Solar panels getting over a thousand times less expensive is already quite extraordinary but this is something else altogether. The chip in a 2026 smartphone has more raw arithmetic capacity than the biggest room sized supercomputer of the 1980s.

On top of this, storage gets cheaper at twenty-five to forty percent per year. The first commercial hard drive shipped by IBM in September 1956 stored five megabytes, weighed over a ton, was moved by forklift, and rented about thirty thousand dollars a month in modern money. A microSD card the size of a fingernail now stores a terabyte for 200 euros. Per dollar, that is roughly 30 million times more storage, plus the fact that you don't need a separate room and several engineers to use it. This fingernail-sized chip holds a thousand feature films, ten thousand books, the entire recorded output of Bach, or all your family photos from the day you were born to the day you die. We now treat this as a normal property of small plastic objects.

Bandwidth -- the speed at which one computer can talk to another -- follows the same shape, only steeper: thirty to fifty percent a year. The dial-up modem on which I send my first emails ran at fourteen kilobits per second: you could practically hear it spell out the letters. A single low-resolution photograph took several minutes, an album of music several hours. Today we stream high-definition video to the back seat of a moving car and we don't even marvel about it anymore. The ethernet in my study runs about a hundred thousand times faster than my first dial-up modem.

On top of that comes the AI. The cost of training and running a large neural network is falling faster than Moore's Law alone could explain, because chip improvements are compounding with algorithmic improvements and software improvements.

Fast forward ten years in the future and you will have a $10 credit card sized Raspberry Pi computer -- holding it's own with current smartphones -- attached to every inverter, battery and solar array. Each appliance will be able to plan and coordinate it's energy use in extremely sophisticated ways, and it will be able to do so basically for free, especially when compared to the cost of the grid, energy supply, and batteries.

We need to turn grid-planning on its head

Currently, fossil fuel and nuclear are still dominant in the way we organize the electricity grid. The fossil fuel lobbyist still have the best access to politicians and regulators. And the grid operators still spend almost all our money on more cables and transformers.

Solar and wind are increasingly taken seriously, but batteries are still seen as a nuisance, grid-forming inverters are hardly on the radar, and when you talk about smart semi-autonomous appliances you might as well save your breath.

Some of this is just generational nostalgia. The European policy frameworks that shape today's conversation were largely written between 1995 and 2015, when wind was the most visible new technology, solar still needed subsidising because it was expensive, batteries were boutique items, and the kind of distributed intelligence we are now talking about was science fiction. The policy frameworks made sense at the time, especially when you don't use learning curves to anticipate how things will change. But the people who wrote them seem unwilling or unable to say clearly that they have not aged well, to put it mildly.

The actual hierarchy of learning rates should invert this whole discussion. Using EVs, batteries and all other flexible devices in the smartest way possible should be the *first* priority. Policy makers or regulators that do not understand that should move out of the way. Making sure every building is run by a grid-forming inverter that is rewarded for keeping the grid stable should be the second priority. Scaling up batteries and enabling V2G should be the third. Scaling up solar and wind the fourth. And only *after* all that is done to the max should we spend money on grid reinforcements and fossil fuel stop-gaps.

The Future is Now

None of this is science fiction. Many home energy management systems are already skilled at optimising self use of solar panels and buying energy optimally, taking into account the weather forecast. They would love to trade with the grid operator too. EV charge controllers do similar arithmetic. Modern domestic solar inverters often are pretty smart. Heat pumps from manufacturers like Vaillant and Daikin learn the household's heating patterns, and modulate accordingly. The hardware is on the wall already. What is missing is not the silicon. What is missing is a system that asks it to do anything interesting.

Other industries are years ahead. A modern car contains more lines of code than a fighter jet and runs models that decide, every few milliseconds, whether the wheel is about to slip on ice. A modern factory floor has sensors on every motor, talking to a local controller that talks to a central one and decides things in microseconds. Even your washing machine can detect, by listening to its own bearings, when it is about to break. The energy system is not going to acquire distributed intelligence as a luxury, or as the result of a heroic government programme. It is going to acquire distributed intelligence because that is what every adjacent industry has been doing for the last two decades.

The problem is systemic inertia

We have technology to use computers, grid-forming inverters and batteries to make the grid incredibly resilient and to use its hardware far more optimally. What's missing is the willingness of increasingly out of touch players like grid operators and their regulators to *use* any of it.

They insist on using the same price everywhere and fixing that price years on advance. We will learn that the scientific literature has agreed decades ago that that's a recipe for problems: you should always try to use prices that reflect reality or there will be no end of waste and gaming the system.

They invest tens of billions per year in the Netherlands alone but somehow can't find the money and motivation to monitor the use of that grid infrastructure over time, let alone disclose the data. The amount of resistance to spending money on intelligence and to sharing data is truly remarkable. Compared to the Internet, the intelligence of the "smart grid" is easily a million times less.

This gap is, as far as I can tell, the most under-appreciated fact in the energy debate. Using a bit more intelligence would easily earn back the investment hundredfold, but what's missing is the willingness to accept that using that intelligence might change things.

I telecom I experienced first hand how the telecom operators didn't want to use Internet technology because it made their customers more powerful and the central grid less powerful. Only when threatened with extinction by cable companies did they change their ways. Grid operators are publicly owned monopolies to they will not face the threat of extinction, but there are a couple of levers we can use to initiate change nevertheless.

But first we have to understand why the grid we inherited is no longer up to the task.